Quantum Fault Tolerance Meets Toom's Contours

Michael Ben-Or David Ponarovsky

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1 Space-Time Graph

So, we connect arrow between $u \in \partial z^{(t)}$ to $v \in \partial z^{(t+1)}$ if there is an edge between u to v in the base graph G_o. Then for any triple $v \in \partial z^{(t+1)}$ and $u, w \in \partial z^{(t)}$ such $\{u, v\}$ and $\{w, v\}$ are in A we connect $\{u, w\}$ by fork. In such way we get a space time graph.

In the shrinking, we have a finite assignment z. After applying the sweep rule $\rightarrow \xi$, we then look at a slice ξ', which is connected by arrows. We find that $\xi' \subset \xi$.

Definition 1.1. Consider a graph $G = (V, A, F)$, where V is a set of vertices, and A, F are sets of undirected edges. For $v \in V$, denote by $\text{pre}(v) = \{u \in V : \{u, v\} \in A\}$, we extend the notion for subsets of vertices $X \subset V$ by $\text{pre}(X) = \bigcup_{v \in X} \text{pre}(v)$. Let $T : V \rightarrow \mathbb{N}$, We say that (G, T) is a time graph if and only if:

1. $\{x, y\} \in A \Rightarrow T(y) = T(x) \pm 1$
2. For $x, y \in V$ $\{x, y\} \in F$ if and only if there exists $z \in V$ such both $\{x, z\}, \{y, z\} \in A$.

We will call to T time-function, to A arrows, and to F forks. Observes that from the second requirement it follows that forks can connect only vertices that have the same time value (set by T).

Definition 1.2 (Time Graph of $C(\mathcal{E})$)**.** For a quantum circuit C and noise $\mathcal{E}$, let the base graph of $C[\mathcal{E}]$ denoted by G_o. For each time t, denote the deviation induced by $\mathcal{E}$ at time t by $\mathcal{D}_t$. We define the history graph $\mathcal{H} = (V_{\mathcal{H}}, E_{\mathcal{H}})$ as follows:

1. The vertices $(v, t) \in V$ if at time t (namely immediately after time t), there is a face (qubit) of $f \in G_o$, such that $v \in f \in \mathcal{D}_t$.
2. We connect edges as follows: if $u_{t-1} = (u, t-1)$ and $v_t = (v, t)$ are both in V, $v_t \notin \mathcal{E}$, and u and v are connected in G_o, then we connect them by an arrow. We name the set of arrow edges A. Moreover, for vertices u_t and w_t, if there exists v_{t+1} such that $\{u_t, v_{t+1}\}, \{w_t, v_{t+1}\} \in A$, then we connect u_t and w_t by a fork.

By definition, the construction $G_{\mathcal{H}}$ is a time-graph.

Definition 1.3 (History, slice.)**.** For a time graph (G, T) and a vertex $u \in V$, let G_u be the subgraph of (V, A) induced by $\{v \in V : T(v) \leq T(u)\}$. We will denote by H_u the 'History of u', which is the set of vertices that are connected to u in G_u. A slice is an equivalence class of the relation $u \equiv v$ if and only if $H_u = H_v$. Notice that the time function has to be constant on a slice, to see this consider vertices x, y with $T(x) > T(y)$, thus $x \notin G_y$ and therefore $x \in H_x/H_y$ So they are belong to different classes. Thus, we can extend the time function and the history to slices and write for a slice K, $T(K)$ and H_K.

Claim 1.4. Consider slices K_1, K_2 such that $T(K_1) = T(K_2)$, then either $K_1 = K_2$ or H_{K_1}, H_{K_2} are disjoint.

Proof. Assume a non-empty intersection, so there exists $z \in H_{K_1} \cap H_{K_2}$ such that for any $x \in K_1$ and $y \in K_2$, there exist arrow-paths $z \to x$ and $z \to y$. Thus, for any $w \in H_{K_1}$, one can concatenate a path $w \to x \to z \to y$ and conclude that $y \in H_{K_2}$. □

Definition 1.5 (The graph of slice)**.** Let K be a slice. The graph of K, denoted by $G_K = (V_K, E_K)$, is the graph whose vertices are slices $R \subset H_K$ with $T(R) = T(K) - 1$. Two vertices are connected by an edge if and only if there is a fork whose ends belong to the slices associated with them. Namely, $S, R \in V_K$ are connected in G_K if and only if there is $\{s, r\} = f \in F$ such that $s \in S$ and $r \in R$.

The connection property is crucial, In our construction, there are not neutral forks, and we will have to find way to overcome it, one way is to take $\arg\min_{u \in \sigma_1, v \in \sigma_2} d(u, v)$ and connect the by fork. More thought should be invested here.

Claim 1.6. G_K is connected.

Proof. Let $R, S \in V_K$ and consider $r, s \in V$ such that $r \in R, s \in S$. Recall that $R, S \subset H_K$ and therefore $r, s \in H_K$, Thus, by definition of the 'History' there exists a path, passing trough arrows only, from r to s in H_K. Consider such a path that is minimal in the number of points x belong to it, with $T(x) = T(K)$. By induction on the number k of such points we prove that R and S are connected in G_K.

1. Base. $k = 0$, In this case, the path from r to s lies within $G_r (= G_s)$. Hence $H_r = H_s$, So $R = S$.
2. Induction step. There exists y on the path from r to s such that $T(y) = T(K)$. Let x be the point which precedes y on the path, and z the point that follows y. Since the values of T at the two ends of an arrow differ by 1, and since y gets the maximal T value in H_K it follows that:
   (a) $\{x, z\} \subset \mathrm{pre}(y) \Rightarrow \{x, z\} \in F$.
   (b) The slices of x and z, say K_x and K_z has time value $T(K_x) = T(K_z) = T(K) - 1$, Namely they are both members of V_K.

Since the path connecting r, x in H_K has $k-1$ vertices with $T(\cdot) = T(K)$, then by the induction hypothesis, R and K_x are connected in G_K, and by similar argument so are K_z and S. On the other hand, $\{x, z\} \Rightarrow \{K_x, K_z\} \in E_K$. Thus, R and S are connected in G_K.

$\square$

Definition 1.7. Let $f : G \hookrightarrow \mathbb{R}^d$ be an embedding that maps each element of $V \cup A \cup F$ to a point in $\mathbb{R}^d$. We say that $\left(G, \mathbb{R}^d\right)$ is a space-time graph if and only if:

1.

2. The point set by f for an edge is the mean of its endpoints' images; that is, for $e = \{x, y\} \in A \cup F$, we have $f(e) = \frac{1}{2}\left(f(x) + f(y)\right)$.

Definition 1.8. Define $\text{pre}_i(v) = \arg\max_{u \in \text{pre}(v)} L_i(f(u))$

Claim 1.9. Let $\mathbf{S} = \{S_1, S_2, .., S_k\}$ where $S_i \subset \mathbb{R}^d$, introduce an edge between S_i and S_j iff $S_i \cap S_j \neq \emptyset$. Assume that $\mathbf{S}$ is connected. Then $\mathbf{Size}(\cup_{i=1}^k S_i) \leq \sum_{i=1}^k \mathbf{Size}(S_i)$.

Proof. It suffices to prove that if $R_1 \cap R_2 \neq \emptyset$, then $\mathbf{Size}(R_1 \cup R_2) \leq \mathbf{Size}(R_1) + \mathbf{Size}(R_2)$. Let $a_{i,j} = \max_{v \in R_j} L_i(v)$. Then for any $w \in R_1 \cap R_2$ we have:

$$\begin{aligned}
\mathbf{Size}(R_1 \cup R_2) &= \sum_i^{d+1} \max a_{i,1}, a_{i,2} = \sum_i^{d+1} a_{i,1} + a_{i,2} - \min a_{i,1}, a_{i,2} \\
&\leq \sum_i^{d+1} a_{i,1} + a_{i,2} - L_i(w) = \mathbf{Size}(R_1) + \mathbf{Size}(R_2) - \mathbf{Size}(w) \\
&= \mathbf{Size}(R_1) + \mathbf{Size}(R_2)
\end{aligned}$$

$\square$

Definition 1.10 (Spanned Set)**.** Consider an embedding $f : \star \hookrightarrow \mathbb{R}^d$, a subset of f's domain $S \subset \star$, and $d+1$ points $x_1, \ldots, x_{d+1}$. We say that $\langle S, x_1, \ldots, x_{d+1} \rangle$ is a spanned set of S if for any $i \in [d+1]$ and $s \in S$, it holds that $L(f(s)) \leq L_i(x_i)$ and in addition there are $v_1, ..v_{d+1} \in V$, not necessary different, that achieve the equalities $L_i(x_i) = L_i(f(x_i))$. We name $x_1, \ldots, x_{d+1}$ the poles of S. Furthermore, we extend the notation to a spanned slice when S is a slice of some time-graph. Notice that saying $\langle S, x_1, \ldots, x_{d+1} \rangle$ is a spanned set means that $\mathbf{Size}(S) = \mathbf{Size}(\{x_1, \ldots, x_{d+1}\})$.

Observe that in the case where the subset S is a slice. Each of these poles can be associated with a vertex of G that gives the equality. If there are multiple vertices whose images by f give equality, associate the point with one of them arbitrarily. We will use the notion of applying functions originally defined over vertices to the poles, referring to the application over the vertices associated with them. For example $\text{pre}(x_i) = \text{pre}(v)$ for the vertex $v \in S$ which satisfies $L_i(f(v)) = L_i(x_i)$ and therefore is associated with the pole x_i. Another example is referring to x_i as a vertex, so when saying 'For $x_i \in V_k$' or 'For $\{x_i, u\} \in F$', we refer to the associated vertex or the edge connecting it to u.

Claim 1.11. Consider a time graph (G, T), an embedding $f: G \to \mathbb{R}^d$, such that.. , Let K be a slice in G, and let $\hat{K} = \langle K, x_1, ..x_{d+1} \rangle$ be a spanned slice of K such that $K \neq H_K$. Then for some integer k there exist spanned slices $\hat{K}_0 = \langle K_0, \cdot \rangle, .., \hat{K}_k = \langle K_k, \cdot \rangle$ and Forks $F_1, .., F_k$ such that:

1. For $i \in [d+1]$ $\text{pre}_i(x_i)$ belongs to the poles $\hat{K}_0, .., \hat{K}_k$.
2. Introduce an edge between $\hat{K}_i$ and $\hat{K}_j$ if and only if one of the Forks $F_1, .., F_k$ consists of a pole of $\hat{K}_i$ and a pole of $\hat{K}_j$. Then the graph formed from $\hat{K}_0, .., \hat{K}_k$ is connected.
3. $\sum_{i=0}^{k} \mathbf{Size}(\hat{K}_i) + \sum_{i=1}^{k} \mathbf{Size}(F_i) \geq \mathbf{Size}(\hat{K}) + 1$

Proof. First Notice that K doesn't contain leaves in (V, A), since a leaf is connected only to itself via arrows, and therefore its 'slice' equals to its 'History'. Thus $\text{pre}_i(x_i)$ is not empty for $i = 1, .., d+1$. Next consider the graph $G_K = (V_K, E_K)$ from definition 1.5. Since $\{x_i, \text{pre}_i(x_i)\}$ is an arrow $\text{pre}_i(x_i)$ belongs to a slice in V_K. for $i = 1, .., d+1$ denote by R_i the slice in V_K that contains $\text{pre}_i(x_i)$.

By claim 1.6 G_K is connected and therefore there exist a minimal collection $\{K_0, .., K_k\}$ of slices form V_K which contains $R_1, R_2, ..R_{d+1}$, and which is connected in G_K. We pick a set of forks $\mathbf{F} = \{F_1, .., F_k\}$ which realizes a spanning tree for this collection of slices. Later we will identify edges from this spanning tree with the forks from $\mathbf{F}$.

For $i = 1, .., d+1$ we set $\text{pre}_i(v_i)$ to be the ith pole of R_i, This assures (1). The set of remaining poles for $\hat{K}_0, .., \hat{K}_k$ will be equal to $\cup_{i=k}^{k} F_i$. This will assure (ii), We will also select poles for $F_1, .., F_k$ to form 'spanned forks' $\hat{F}_1, \cdots, \hat{F}_k$ in such way that:

$$\sum_{i=0}^{k} \mathbf{Size}(\hat{K}_i) + \sum_{i=1}^{k} \mathbf{Size}(\hat{F}_i) \geq \sum_{i=1}^{d+1} L_i\left(\text{pre}_i(v_i)\right) \geq \mathbf{Size}(\hat{K}) + 1$$

This will assure (3).

Let us denote by $J_0, .., J_{2k}$ the enumerated union $K_0, ..K_k, F_1, .., F_k$. Observes that for any choice of $(d+1)(2k+1)$ elements $z_0^1, z_0^2, z_0^3, .., z_0^{d+1} .., z_{2k}^{d+1}$ in $\mathbb{R}^d$, such that $z_j^i \in J_j$ it holds that:

$$\sum_{i=0}^{k} \mathbf{Size}(\hat{K}_i) + \sum_{i=1}^{k} \mathbf{Size}(\hat{F}_i) \geq \sum_{j,i} L_i(z_j^i) \tag{1}$$

Later, we are going to take the points $\{z_j^i\}$ from the intersection of J_j's. For convenience, let us assume that for the index subset $X \subset [2k] \cup \{0\}$, the intersection $\bigcap_{j \in X} J_j$ returns a single point. If there is more than one, then decide on the point in a way that all the chosen points simultaneously satisfy that for any index subsets Y, X, such that $X \cap Y$ and they both introduce a non-trivial intersection $\bigcap_{j \in Z} J_j$: $Z \in \{X, Y\}$, then:

$$\bigcap_{j \in X} J_j = \bigcap_{j \in Y} J_j$$

Now consider a non-trivial maximal intersection $\bigcap_{j\in X} J_j$, the intersection is maximal in the sense that for any $j' \notin X$ it holds that .$\bigcap_{j\in X\cup\{j'\}} J_j = \emptyset$. We claim that for any i and for any such X, there is $t \in X$ such that J_t is the successor of J_j on the path to R_i for any $j \in X/\{t\}$. Assume not. Let $x, y \in X$, and denote the paths from J_x, J_y to the ith pole by $\langle J_x, J_{x_2}, \ldots, R_i\rangle$, $\langle J_y, J_{y_2}, \ldots, R_i\rangle$, such that $J_y \neq J_{x_2}$. If $y = x_2$, $y_2 = x$, or $x_2 = y_2 \in X$, then pick other x, y (if there is no such, we get an immediate contradiction). In that case, $\langle J_x, J_y, J_{y_2}, \ldots, R_i\rangle$ is a path from J_x to R_i which is different from $\langle J_x, J_{x_2}, \ldots, R_i\rangle$. Therefore, there are at least two paths from J_x to the ith pole, namely, there is a cycle in contradiction to the minimality of $K_0, \ldots, K_k$.

We will choose the z_j^i by the following process. Pick a z_j^i that has not been set yet. If $J_j = R_i$, then set $z_j^i \leftarrow \mathrm{pre}_i(x_i)$ - we call this a type-1 assignment. Otherwise, consider the path from J_j to R_i (by connectivity, it has to be such a one, and by minimality, there is exactly one). Let J_t be the successor on the path, then pick $z_j^i \in J_j \cap J_t$ - similarly, we call this assignment a type-2 assignment. Since any type-2 assignment is associated with a non-trivial intersection, and any non-trivial intersection induces a type-2 assignment for all i's (since for all i's, one of the J_j on its support is a successor of the rest on their path to the ith pole), we have that:

$$\begin{aligned}
\sum_{j,i} L_i(z_j^i) &= \sum_{j,i,z_j^i\in\text{type-1}} L_i(z_j^i) + \sum_{j,i,z_j^i\in\text{type-2}} L_i(z_j^i) \\
&= \sum_{i}^{d+1} L_i\left(\mathrm{pre}_i(x_i)\right) + \sum_{X:\ \bigcap_{j\in X} J_j\neq\emptyset} (|X|-1) \sum_{i}^{d+1} L_i\left(\bigcap_{j\in X} J_j\right) \\
&= \sum_{i}^{d+1} L_i\left(\mathrm{pre}_i(x_i)\right)
\end{aligned}$$

Plugging it into eq. (1) gives the desired. $\square$